

Practice Quiz: Body Science — Module 01: Systems

Name: ____ Date: ____

Part A: Multiple Choice

1. Your body is a system. What makes it a system rather than just a collection of parts?

A) It has a lot of cells B) Its parts (organs, tissues, cells) are connected and work together to keep you alive C) It is made of water D) It exists inside your skin

1. The "boundary" of a cell is called the cell membrane. What does this boundary do?

A) It blocks everything from entering or leaving B) It controls what comes in and what goes out, keeping the cell's insides stable C) It has no function D) It destroys anything that touches it

1. Your digestive system breaks down food, your circulatory system carries nutrients through blood, and your cells use those nutrients for energy. This is an example of:

A) Three systems that have nothing to do with each other B) Systems within a larger system — they are nested and connected C) Only one system D) Systems that work against each other

1. What is homeostasis?

A) A disease B) Your body's ability to keep its internal conditions (like temperature and blood sugar) stable and predictable C) A type of exercise D) Something that only happens when you sleep

1. When you get a fever during an illness, your body's temperature system is:

A) Broken beyond repair B) Temporarily changing its set point — adjusting its prediction of what temperature it should be to fight the infection C) Ignoring the illness entirely D) Shutting down permanently

1. Which is the best example of a "ripple effect" in your body?

A) You breathe in and nothing else happens B) You eat a sugary snack, your blood sugar spikes, your pancreas releases insulin, and your cells absorb the sugar C) Your hair grows by itself D) You blink once

1. Your nervous system and your immune system both protect you, but in different ways. Together they show that:

A) Body systems are isolated and never interact B) Different systems communicate and coordinate to handle challenges the body faces C) Only one system matters at a time D) The body has no overall purpose

Part B: Short Answer

1. Pick one body system (nervous, digestive, respiratory, etc.) and describe its **parts**, how they are **connected**, and what the system's **purpose** is. What happens when one part of that system stops working properly?

2. Explain what **homeostasis** is using the example of body temperature. What does your body "predict" it should be? What happens when the actual temperature doesn't match (like when you have a fever or you're in a cold room)?

3. Your body is a system made up of smaller systems (organs, cells), and it also exists inside larger systems (your family, your environment). Describe this "nesting" of systems. How does something happening at the environmental level (like cold weather) create a ripple effect down to the cellular level?